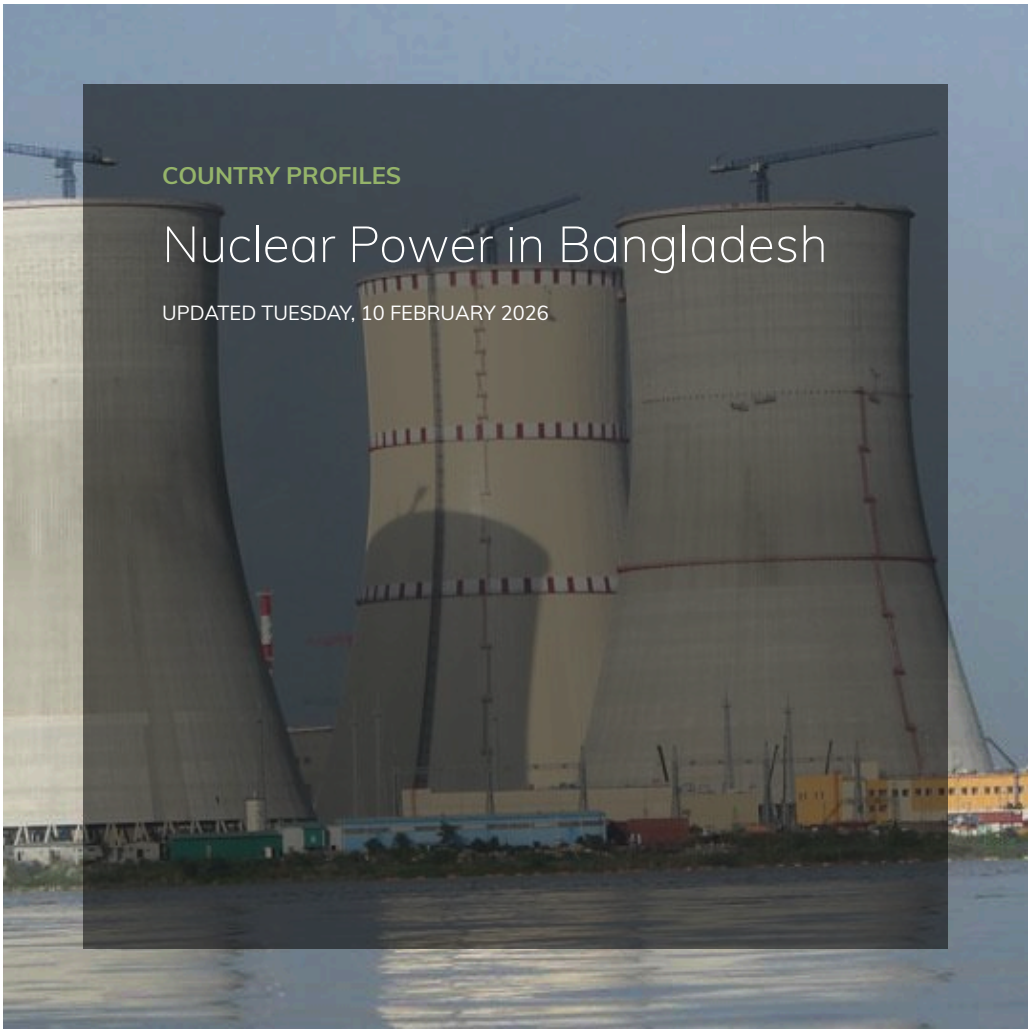

COUNTRY PROFILES

Nuclear Power in Bangladesh

UPDATED TUESDAY, 10 FEBRUARY 2026

- Bangladesh started construction of its first nuclear power reactor, Rooppur 1, in November 2017. The unit is expected to begin operation in 2026.
- Construction of the second unit at Rooppur commenced in July 2018.
- The country has a rapidly increasing power demand and is aiming to reduce its dependence on natural gas.

Privacy - Terms

0	2	0
OPERABLE REACTORS	REACTORS UNDER CONSTRUCTION	REACTORS SHUTDOWN
0 MWe	2,160 MWe	0 MWe

Electricity sector

Total generation (in 2023): 101 TWh

Generation mix: natural gas 66.4 TWh (66%); oil 21.9 TWh (22%); coal 10.7 TWh (11%); solar 1.3 TWh; hydro 0.6 TWh.

Import/export balance: 10.4 TWh imports, no exports.

Total consumption: 97.6 TWh

Per capita consumption: c. 500 kWh in 2023

Source: International Energy Agency and The World Bank. Data for year 2023.

Electricity demand is rising rapidly (about 7% per year) and capacity was 28.9 GWe as of November 2025, up from 5 GWe in 2009[1].

According to the World Bank, about 99% of the population had access to electricity by 2023, though those relying on the grid experience frequent power cuts. Some 5% of government expenditure is being allocated to 'power and energy' (see figures published by the Bangladesh government's Power Division).

A long-term plan was made in the Power System Master Plan (PSMP) 2016[2], an updated version of the PSMP 2010. The plan aimed for generation capacity of 23 GWe in 2020, 40 GWe in 2030, and 60 GWe in 2041. The first contribution from nuclear was initially expected in 2023, and the plan shows approximately 7 GWe nuclear capacity by 2041. The PSMP 2016 was superseded in 2023 by the Integrated Energy and Power Master Plan (IEPMP), which the interim government is revising.

Nuclear power plants

Under construction reactors in Bangladesh

Reactor name	Model	Process	Gross capacity (MWe)	Construction start
Rooppur 1	VVER V-523	PWR	1200	2017-11-30
Rooppur 2	VVER V-523	PWR	1200	2018-07-14

Source: World Nuclear Association

Rooppur

Building a nuclear power plant in the west of Bangladesh was first proposed in 1961. The Rooppur (or Ruppur) site in Pabna district about 160 km northwest of Dhaka on the banks of the Ganges river

was selected in 1963 and land was acquired. The government gave formal approval for a succession of plant proposals, then after independence a 125 MWe nuclear power plant proposal was approved in 1980, but not built.

With growth in demand and grid capacity since then, a much larger plant looked feasible, and the government in 1999 expressed its firm commitment to build a plant at Rooppur. In 2001 it adopted the Nuclear Power Action Plan and in 2005 it signed a nuclear cooperation agreement with China.

In 2007 the Bangladesh Atomic Energy Commission (BAEC) proposed two 500 MWe nuclear reactors for Rooppur by 2015, quoting likely costs of US$ 0.9-1.2 billion for a 600 MWe unit and $1.5-2.0 billion for 1000 MWe. In April 2008 the government reiterated its intention to work with China in building the Rooppur plant and China offered funding for the project. The International Atomic Energy Agency (IAEA) approved a technical assistance project for the Rooppur nuclear power plant to be initiated between 2009 and 2011, and it then appeared that a 1100 MWe plant was envisaged.

Russia, China and South Korea had all earlier offered financial and technical help to establish nuclear power in the country, and in March 2009 Russia made a formal proposal to build a nuclear power plant. In April 2009 the government approved the Russian proposal to build a 1000 MWe AES-92 nuclear plant at Rooppur for about $2 billion, and in May 2009 a bilateral nuclear cooperation agreement was signed by the two countries. A year later the proposal had expanded to include two such reactors by 2017.

In May 2010 an intergovernmental agreement was signed with Russia, providing a legal basis for nuclear cooperation in areas such as siting, design, construction and operation of power and research nuclear reactors, water desalination plants, and elementary particle accelerators. Other areas covered included fuel supply and wastes.

An agreement with Rosatom was signed in February 2011 for two 1000 MWe-class reactors to be built at Rooppur for the Bangladesh Atomic Energy Commission (BAEC). Rooppur is close to a HVDC link with India and on the route of a planned 600 kV HVDC link running up the western side of the country. Another intergovernmental agreement was signed in November 2011 for the project to be built by AtomStroyExport. In 2014 Moscow AEP said that the plant would be an AES-2006 with a version of its V-392M reactors, with Novovoronezh II being the reference

plant. Turbine generators (high speed) are expected to be from Power Machines.

A nuclear energy bill was introduced into parliament in May 2012, which outlined the establishment of the Bangladesh Atomic Energy Regulatory Authority. Parliament was told that 5000 MWe of nuclear capacity was envisaged by 2030, and that a second plant would be built in the south once Rooppur is operating.

An intergovernmental agreement for provision of a $500 million Russian loan to finance engineering surveys on the site, project development and personnel training was signed in January 2013. The $500 million loan will be repaid in 12 years with five years' grace period. In June and October 2013, AtomStroyExport signed contracts with BAEC to prepare documentation related to the construction and environmental impact assessment for the Rooppur plant, as well as providing for necessary engineering studies, including site preparation and detailed design documentation. AtomStroyExport said that this represented a transition to long-term cooperation. Site works started in October 2013.

A $190 million agreement for major site works to prepare for first concrete was signed in June 2014, and BAEC authorized this in September, to be completed by 2016.

In February 2015 Orgenergostroy, a Russian company, won an $18 million tender to carry out an engineering survey, perform environmental monitoring and develop project documents for the plant. In December 2015 Russia’s InterRAO Engineering won a RUR 2.1 billion ($30 million) contract from Atomstroyexport for the concrete foundations.

Site works were 80% complete in April 2016, and a site licence was issued by the Atomic Energy Regulatory Authority in June 2016.

The project management company is JSC NIAEP, part of AtomStroyExport. The main BAEC contract with AtomStroyExport signed in December 2015 amounts to $12.65 billion, including the first few years' fuel, with Russia financing 90% of the cost at an interest rate of Libor plus 1.75%, capped at 4%, repayable in 28 years with 10 years' grace period. In June 2016 the cabinet approved a draft agreement for Russia to provide an $11.38 billion credit facility for 90% of the project, and the agreement was signed in July. Bangladesh needs to pay its 10% of the contract value in advance. The Russian Bank for Development and Foreign Economic Affairs and Bangladesh's state-owned Sonali Bank would work as authorized banks for the state credit.

Construction of the first Rooppur unit commenced in November 2017, with commissioning initially expected in 2023. The second unit commenced construction in July 2018, and was initially expected to be completed in 2024. Commissioning of unit 1 is now expected in 2026. It is a turnkey project, and Rosatom will maintain the plant for the first year of operation before handing over to the BAEC. The IAEA continues its close involvement with the project. In December 2021 Rosatom reported that it had installed all four steam generators for unit 1. Dummy fuel assemblies were loaded into the unit 1 reactor in September 2024. Cold and hot functional tests were completed in 2025.

Rosatom is undertaking some staff training, and by the end of 2019, more than 500 Bangladeshis had undergone an internship at Russia's Novovoronezh II nuclear power plant.

Earlier in May 2015 Bangladesh asked India to help with training its nuclear staff and in April 2017 BAEC announced the appointment of India's Global Centre for Nuclear Energy Partnership (GCNEP) as the consultant for construction and operation of the Rooppur project. GCNEP is a Department of Atomic Energy R&D facility strongly supported by Russia and designed to strengthen India's collaboration internationally. In March 2017 the IAEA agreed to provide staff for the centre and use it for training professionals throughout the region.

In October 2022 the government announced that the construction of Rooppur was running approximately one year behind schedule due to issues stemming from the Covid-19 pandemic and Russia's invasion of Ukraine. The same month, the Russian government issued a mandate to Bangladesh's government to repay the loans provided for the construction of the plant in Russian rubles instead of US dollars. The revised schedule, agreed in June 2025, set unit 1 commercial operation for December 2026 and unit 2 for December 2027.

Earlier in April 2023 as a result of sanctions on Russia making payment in US dollars not possible, the Bangladesh government opted to settle a reported $318 million pending repayment in Chinese yuan.

In August 2024, following a student-led uprising, Prime Minister Sheikh Hasina resigned and an interim government was formed under Muhammad Yunus. The interim government pledged to continue the Rooppur project, and in February 2025 agreed with Rosatom to extend the credit utilization period until the end of 2026. A proposal to increase the project cost by about 23%, mainly

due to depreciation of the taka against the dollar, and to extend the overall project completion date to June 2028, is under review.

Second nuclear power plant

In June 2014 the government and BAEC invited the Japan Atomic Energy Agency to explore the possibility of building a second 2000 MWe nuclear power plant in the south of the country. Site selection focused on several islands in the Bay of Bengal, notably Mazher Char in Barguna, Muhurir Char in Feni, Boyar Char in Noakhali and Gangamati in Patuakhali. Four other potential sites are in Khulna, close to the Sundarbans. In August 2018 it was reported that the site would be near a new deep-water port. In October 2021 it was reported that site selection continued, with search for competent geological substrate.

China's Dongfang Electric Corporation (DEC) has expressed interest in building the second power plant, with China Nuclear Engineering & Construction Corp (CNEC). In April 2019 the sites under consideration were Gangamati in Patuakhali district, Mazher Char in Barguna (both south-central), Boyar Char in Noakhali, and Muhurir Char in Feni (both east), with Gangamati favoured. In April 2024, Prime Minister Hasina and Rosatom Director General Likhachev discussed the possibility of building two additional units at the Rooppur site.

Reactors proposed in Bangladesh

Reactor name	Location / site	Type	Gross capacity (MWe)
?	Gangamati?	VVER?	2x1200

Fuel cycle

All fuel for Rooppur is being provided by Rosatom. A contract between Rosatom's fuel company TVEL and BAEC was finalized in August 2019. All used fuel is to be repatriated to Russia, in line with standard Russian practice for such countries. A draft agreement on used fuel was signed in March 2017, totalling about 22.5 t/yr from each reactor (42 fuel assemblies, each with 534 kg of fuel). A further agreement for repatriation of used fuel for reprocessing was signed in August 2017.

BAEC is doing some uranium exploration focusing on Tipam sandstone near the Indian border in the north.

Research

Bangladesh's development strategy (Vision 2041) aims for the country to reach high-income status by 2041, in large part by emphasis on its science and technology sector to drive economic growth. The ministry of science and technology (MoST) estimated in 2014 that $6.2 billion would be needed in the next decade to achieve the goals of Vision 2021. The Science and Technology Act 2010 is helping to boost this, and MoST is now allocating over $150 million per year to nuclear technology development as gas reserves become depleted.

The country has had a Triga 3 MW research reactor operational since 1986.

Regulation, safety and non-proliferation

The Nuclear Power Plant Act 2015 set up the Nuclear Power Company of Bangladesh (NPCB) to run the plant, though ownership remains with the BAEC. A new Atomic Energy Commission bill was passed by parliament in 2017, updating 1973 legislation.

In February 2012 the MoST signed an agreement with Russia's Rostechnadzor related to regulation and safety "and the provision of advisory support to the Bangladesh Atomic Energy Regulatory Authority (BAERA) on regulation, licensing and supervision." Staff will be trained in Russia.

In May 2016 an IAEA review following up a 2011 IAEA integrated nuclear infrastructure review (INIR) reported that responses to the INIR were in train.

In April 2017 BAERA signed an agreement with India's Atomic Energy Regulation Board (AERB) on the exchange of technical information and cooperation in the regulation of nuclear safety and radiation protection.

Three nuclear energy agreements between Bangladesh and India were signed in April 2017. These comprise a nuclear cooperation agreement between Bangladesh's Ministry of Science and Technology and India's Department of Atomic Energy (DAE), a regulatory cooperation agreement between BAERA and India's AERB, and an agreement between BAEC and DAE's GCNEP on "cooperation regarding nuclear power projects in Bangladesh" (see above).

Non-proliferation

Bangladesh is a party to the Non-Proliferation Treaty (NPT) and has had a safeguards agreement in force with the IAEA since 1982, and an Additional Protocol in force since 2001.

Notes & references

References

1. Bangladesh Ministry of Power, Energy and Mineral Resources [Back]

2. Power System Master Plan 2016, Bangladesh Ministry of Power, Energy and Mineral Resources (September 2016) [Back]